

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	549038
Award Program	Polliantor Health Fund
Project Title	Investing in our future: training taxonomists and wild bee conservation
Grant Start Date	01/22/2018
Grant End Date	01/21/2023
Total Grant Budget	\$546542
Total FFAR Budget	\$ 1093084
Matching Funder(s)	University of New Hampshire
Final Report Date	01/29/2024
Project Director/Principal Investigator Information	
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General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.

University of New Hampshire Durham New Hampshire &
York University, Toronto, Canada

- 1.2. How many new jobs were created because of this grant funding?

8

- 1.3. How many jobs were maintained because of this grant funding?

2

- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

No

- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

No

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

Bees are critical for not only agricultural crop pollination but also greenspace, park and garden sustainability. While most research focuses on commercial and non-native species there is a growing appreciation for the need to protect native and unmanaged pollinators. The impediments to such work include a lack of taxonomic expertise and ongoing biodiversity conservation surveys to classify the wild bee communities in the US and to determine suitable habitat as well as those in need of restoration. This study met both aims by training future taxonomists and providing taxonomic tools for national and international improvements and ease to identify native species. In addition, through long term biomonitoring of wild bees across land use practices we were able to determine new state records as well as range expansions for native as well as exotic species. Taken together, these tools will future proof wild bee conservation efforts by providing critical tools for identifying bees locally but also providing restoration guidelines in terms of pollinator habitat and best practices moving forward. A public field guide was produced with online resources to connect science communication with farmers, gardeners and the general public.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

The key findings of this study were the local and land use features supporting bee populations in the north east. We found that intermediate land use intensity generally supports the most diverse and abundant wild bee communities. Intermediate land use includes pastures and farms using organic practices and lighter mowing practices vs. heavy mowing and conventional pesticides found in our high land use intensity agricultural sites. Further this work found that bees are most abundant and diverse along forest margins and become less common and diverse going into the forest canopy and into open fields. This suggests the farm and forest margins are ideal habitats for wild bees providing a mix of broken stems and bare soil as well as floral patches as ideal habitat for a wide range of wild bee species. Future land use should prioritize margins and reduced land use intensity to promote wild bee habitat and sustainable pollinator populations moving forward. These findings can be readily applied across New England where farms are typically small scale and abutting forested regions. This habitat is common and readily maintained in this region as long as small scale farms and forest patches are maintained and there is no large scale deforestation or large scale land clearing and massive agricultural monocultures introduced across the region.

- 2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and classified in one of the following ways:
- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
 - b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
 - c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

The research impact of this project were three-fold. The potential outcomes to U.S. agriculture and food systems include encouraging farmers and gardeners to plant or maintain mixed crops and gardens as well as embracing edges in forest and farmland to sustain wild bee populations. Second the intermediate outcomes include improved landscape design and

master gardener and school group education through the use of our wild bee field guides and floral planting recommendations. Thirdly, the end outcomes have resulted in training the next generation of wild bee taxonomists, improving wild bee education and outreach and connecting with regional stakeholders including conservation authorities and land use managers to improve pollinator friendly practices moving forward. Our field guide resulted in 2000 copies freely disseminated to school groups, beekeepers, farmers, master gardeners, and conservation authorities across New England. In addition the online pdf version of this book has resulted in over 5,000 engagements online and 500 additional downloads across North America.

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

Aim I. Train future pollinator taxonomists and contribute important taxonomic keys

Over the course of this project we trained 2 undergrads (Hall and Jacobson) and 2 graduate students (Odanaka and Mathiasson) in wild bee identification. Additionally, 2 undergrads assisted with insect identification, imaging and curation of the insect collections. Field collection of wild bees and taxonomic identifications were completed and published (see full citations for publications in Section 3 below) with all data made freely available online in digital repositories (for specific url links see Section 4 below). This milestone was complete in March 2023 and we can report a 100% completion of this aim. Four key publications resultant from this aim include Gotelli et al. 2023 *Methods in Ecology and Evolution*; Odanaka et al. 2022 *Molecular Phylogenetics and Evolution*; and Mathiasson and Rehan 2020 & 2019 *Insect Conservation and Diversity*.

Aim II. Model habitats and interaction networks to evaluate bee stressors

For this aim we hired a postdoc to conduct geospatial and ecological plant-pollinator interaction modeling of wild bee declines. Dr. Sabine Nooten evaluated multiple interacting stressors by combining farm and forest data from field collected specimens to model ecological, geospatial, and bee population characteristics. This work resulted in five publications Nooten and Rehan 2022 *Agricultural and Forest Entomology*; Nooten and Rehan 2020 *Biodiversity Conservation*; Nooten et al. 2020 *Northeastern Naturalist*; Nooten et al. 2020 *Agricultural and Forest Entomology*; Nooten and Rehan 2019 *Ecological Entomology*. This aim was completed in 2022 with the fifth and final publication of this work.

Aim III. Develop integrated educational resources for the bees of the New England region

We produced the first comprehensive guide to the wild bees of eastern North America led by graduate student (Odanaka), undergraduate (Hall), postdoc (Nooten), and myself. The 'Wild Bees of New England: guide to common pollinators and flowers' book is 120 pages, featuring over 150 bee images taken by the Rehan lab, a historical overview of bees and pollination ecology in the region, summaries of all major bee families, genera, and common species to

the region. All 2000 copies of this field guide have been disseminated with the online version exceeding 5,000 visits and 20,000 page views. This aim was completed in 2020 with the production of the field guide but fully excuted and all deliverables disseminatated in 2022. This aim is 100% complete.

- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

No

- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

For Aim 1 we used museum data to determine how contemporary and historic data can be used to infer conservation status of a wide range of taxa include fish, plants, birds, mammals butterflies and bees. I contributed the bee data from our collection including extensive contemporary sampling throughout this project as well as 150 years of digitized data from the UNH insect collection. This bee data resulted in three major publications including Gotelli et al. 2023 Methods in Ecology and Evolution contributing to this larger cross taxa comparison of conservation status using museum specimens. Here we found that museum holdings are reflective of contemporary populations and distributions and thus future studies should characterize museum records as and important reflection of the biodiversity and conservation status of a wide range of species in a region. Further we specifically analyzed these data for bees resulting in two publications by Mathiasson and Rehan in 2019 and 2020 in the journal Insect Conservation and Diversity. In the 2019 publication we identified that many wild bee species are in decline not just the typically regarded bumble bees and so a wider range of bees including sweat bees, leaf cutter bees, cleptoparasites and mason bees are all of conservation concern. In the 2020 publication we examined the relationship of these declining bee species to plant species extinction in the region and found that wild bee declines were correlated with local wildflower extraptions. Here further emphasizing the need for appropriate wild flowers to sustain wild bee populations and specifically highlighting regionally extinct species in need of restoration. Our hope is that if the wildflower species return and can become established perhaps then also the bees can return and repopulate this region.

For Aim 2 we examined bees across a range of farm and forest practices found in New England and characterized the effects on ground nesting bees across seasons in Nooten and Rehan 2022 Agricultural and Forest Entomology; as well as across crop species Nooten et al. 2020

Northeastern Naturalist; 2022 Agricultural and Forest Entomology. Here we found that sweat bees are impacted by seasonal variation in rain fall and land use. Overall populations fluxuate among season and years and these long term data are needed to assess the prevalence and stability of wild bee populations. Crops like apple and blueberry which are key commodities in the north east and specifically in New England support a wide range of wild bee species and bloom times are well aligned with many mason and leaf cutting bee species as well as bumble bee and sweat bee populations. The prevalence of ground nesting and solitary bees emphasizing the need to reduce mowing and pesticide use as well as no need for introduced honey bees to support local agriculture.

For Aim 3 we developed the field guide for wild bees in New England and this was a tremendous success. Not only was it a pleasure to compile this information and develop the 150 photographs of local wild bees and flower species, but also the broader interest and impact has been most meaningful. While the above aims have been communicated to stakeholders through scientific publications and our outreach website www.wildbeeguide.com, this aim was a very tangible deliverable in the form of a field guide and printed book that was well received by school groups, farmers, gardeners, families, conservation authorities, beekeepers, and the broader community. Here we highlight many general information on the diversity of kinds of wild bees, seasonal wildflowers native to the region and provide recommendations like reduced pesticides, mowing and leaf/stick removal to improve wild bee habitat and sustainability. As noted above, the 2000 copies were all delivered to welcoming stakeholders and became a feature of many world bee day and pollinator appreciation day events. We will continue to host this information on our website and the public engagement of this aim has been tremendous.

- 2.7. Discuss efforts taken to ensure the approach is scientifically rigorous.
Include approaches taken to ensure robust and unbiased results.

For the Aim 1 and 2 scientific components, we were diligent to work with expert taxonomists to ensure our identification and resources were correct. This included sending voucher specimens to national and international experts to confirm identifications as well as extensive photography and email correspondence. All data are digitized and made publically available on the GBIF repository (linked in section 4 below) for maximal utility to the broader scientific and conservation community.

For Aim 3, we fact checked all photos and life history data against published natural history for each species and also asked two leading experts to read the book prior to publication. Sam Droege (USGS Maryland) and Joan Milam (UMass Amherst) both generously and thoroughly peer reviewed this work as local experts in wild bee taxonomy, natural history and wildflower ecology. They provide helpful feedback and suggestions to ensure all data were accurate and of greatest impact to the general public and scientific community.



- 2.8. List key stakeholders that could be served by results of this project.

Farmers, gardeners, community members including private citizens for home and yard improvements, as well as land use managers and conservation partners such as US Fish and Wildlife, Xerces Society, Pollinator Partnership and Bee City USA.

- 2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

Nothing to report

- 2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

None

- 2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

This project trained 8 students in my lab. All 2 postdocs are continuing onto scientific careers as further postdoctoral fellows and one is now an Assistant Professor. The 2 graduate students are also remaining in science with one now postdoctoral fellow and the other working in science education and communication at a natural history museum. The undergraduates have continued on in sciences as well. One is now in graduate school for veterinary medicine, two are in graduate school for ecology and evolution and the fourth employed by a local conservation authority. The skills learned in science communication, taxonomy, ecology and conservation have all been vital to their resultant continuing education and career choices.

- 2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

This projects supported two postdoctoral researchers, two graduate students and four undergraduate research assistant.

3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

Scholarly publications, field guide to wild bees of New England, outreach website (student names in bold font below; PI underlined)

- 3.2. Please provide a list of citations for the information products produced during the project.

Odanaka KA, Hall J, Nooten SS, Rehan SM (2018) *Wild bees of New England: a field guide to common pollinator and flowers*. Foundation for Food and Agriculture Research. 120 pp.

- 3.3 Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

Gotelli NJ, Booher DB, Urban MC, Ulrich W, Suarez AV, Skelly DK, Russell DJ, Rowe RJ, Rothendler M, Rios N, Rehan SM, Ni G, Moreau CS, Magurran AE, Jones FAM, Graves GR, Fiera C, Burkhardt U, Primack RB (2023) Estimating species relative abundances from museum records. *Methods in Ecology and Evolution*. 14:431-443

Odanaka KA, Branstetter MB, Tobin KB, Rehan SM (2022) Phylogenomics and historical biogeography of the cleptoparasitic bee genus *Nomada* (Hymenoptera: Apidae) using ultraconserved elements. *Molecular Phylogenetics and Evolution*. 170:107453.

Nooten SS, Rehan SM (2022) Effects of land use type and seasonal climate on ground nesting wild bees. *Agricultural and Forest Entomology*. 24:229-236.

Nooten SS, Odanaka KA, Rehan SM (2020) Effects of farming practice and seasonal phenology on wild bees in blueberry orchards. *Northeastern Naturalist*. 27:841-860.

Mathiasson ME, Rehan SM (2020) Wild bee declines linked to plant-pollinator network

changes and plant species introductions. *Insect Conservation and Diversity*. 13:595-605.

Odanaka KA, Rehan SM (2020) Wild bee distribution near forested margins is dependent on successional stage. *Forest Ecosystems*. 7:26.

Nooten SS, Odanaka KA, Rehan SM (2020) Characterisation of wild bee communities in apple and blueberry orchards. *Agricultural and Forest Entomology*. 22:157-168.

Nooten SS, Rehan SM (2020) Historical changes in bumble bee body size and range shift of declining species. *Biodiversity and Conservation*. 29:451-467.

Odanaka KA, Rehan SM (2019) Impact indicators: effects of land use management on wild bee functional diversity. *Agriculture Ecosystem and Environment*. 286:106663.

Nooten SS, Rehan SM (2019) Agricultural land use yields reduced foraging efficiency and unviable offspring in the wild bee *Ceratina calcarata*. *Ecological Entomology*. 44:534-542.

Mathiasson ME, Rehan SM (2019) Status changes in the wild bees of northeastern North America over 125 years revealed through museum collections. *Insect Conservation and Diversity*. 12:278-288.

Tucker EM, Rehan SM (2019) Wild bees (Hymenoptera: Apoidea) of the Ossipee Pine Barrens. *Northeastern Naturalist* 26:379-391.

Tucker EM, Rehan SM (2018) Farming for bees: annual variation in pollinator populations across agricultural landscapes. *Agricultural and Forest Entomology* 20:541-548.

- 3.3. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

Field guide to general wild bee biology: <http://www.wildbeequote.com/>

- 3.4. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

NO

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#0	#0	
Copyrights	#0	#0	

Trademarks	#0	#0	
Inventions	#0	#0	

- 3.5. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

No

- 3.6. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that others can access and re-use these data in the future?

All books were made freely available in print copies but also freely available online across lab, outreach and emailed link attachments for maximal dissemination. All papers were published in open access journals when possible and also made freely available on lab website, research gate and to colleagues and researchers upon request via email.

4. Data Management

- 4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imageries). Please list the data generated for the award.

All genetic data is deposited in NCBI Bioprojects portal – free open access for all sequence data.

All museum record data is deposited to GBIF archives for open access records searchable and downloadable from their website.

- 4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

No

- 4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if



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necessary, in a compressed folder.

All data available on the web via the following links:

National Center for Biotechnology Information (NCBI) Bioproject sequence data:

<https://www.ncbi.nlm.nih.gov/bioproject/?term=PRJNA705977>

Global Biodiversity Information Facility (GBIF) museum record data:

<https://www.gbif.org/publisher/154da9ab-c010-422d-8fef-dbc54d10a3c6>



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Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: _____ Sandra Rehan _____

PI Signature: _____ *Sandra Rehan* _____ Date: _____ January 29, 2024 _____

Authorized Signing Official (ASO) Name: _____

ASO Signature: _____ **Kelly J. Marti** _____ Date: _____

Digitally signed by Kelly J. Marti
Date: 2024.02.01 12:48:25 -05'00'